

Shakibul Chowdhury

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M.S. Candidate, Physics, City College of New York

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Current Position

M.S. Candidate in Physics, City College of New York (expected Fall 2025)

Research Areas

- General Relativity and Compact Objects (Neutron Stars, Black Holes)
- AGN & Accretion Physics and Polarimetry
- High-Mass X-ray Binaries (Cyg X-1)
- Black-hole Spin Diagnostics (Polarimetric Methods)
- Neutron-star EOS and Tidal Deformability
- Observational Astrophysics (Binary-star Photometry)

Education

City College of New York (CUNY)

M.S. Physics, expected Fall 2025

Advisors: Prof. Timothy Boyer, Prof. Joshua Tan

Stony Brook University (SUNY)

B.S. Physics and Astronomy, 2020

Undergraduate research with Prof. James Lattimer (Neutron-star EOS)

Research Projects

Independent Study: Schwarzschild Light Bending and X-ray Polarization of Cyg X-1
(Fall 2025)

Mentor: Prof. Joshua Tan (CUNY/AMNH)

- Implemented exact geodesic bending integrals, validated with Beloborodov mapping.
- Modeled disk-integrated Stokes (I, Q, U); analyzed polarization degree and angle trends.

GR Polarimetric Diagnostics of Black-hole Spin (Kerr) (2025-present)

Reader: Prof. Timothy Boyer (CCNY)

- Ray tracing in Kerr spacetime with polarization basis transport.
- Defined spin-sensitive metrics: EVPA winding, radial $p(r)$, and rotation measure statistics.

Binary-star Photometry Project (Great Basin Observatory) (2025 - present)

Mentor: Jerry Hilburn

- Candidate selection via Gaia DR3; light-curve extraction and variability screening.

Neutron-star EOS Studies (2019–2020)

Mentor: Prof. James M. Lattimer (SBU)

- Parsed EOS tables; produced $M-R$ curves and tidal deformability $\Lambda(M)$.

Science Communication & Outreach

Science Communication Intern, AMC (2017)

- Produced educational content and outreach programming for public astronomy events.

Outreach Intern, Astrocamp (2022)

- Led astronomy activities for K–12 students, including telescope observing and classroom presentations.

Teaching Assistant / Lab Instructor, CCNY Physics (2023–2025)

- Supported PHYS 207/208 labs; guided students through experimental design and error analysis.

Regents Prep Tutor, Truman High School (2022–2025)

- Prepared Algebra 2 and Geometry students for Regents exams; authored custom problem sets.

Scholarships & Awards

- Dean's List (Stony Brook University, Fall 2016, Spring 2020)

- Dean's List (CCNY, Spring and Fall 2024 and 2025)

Presentations

- Planned: CCNY Research & Creativity Conference (2025) — two posters (Cyg X-1; GR spin diagnostics)

Manuscripts in Preparation

- *General Relativistic Polarimetric Diagnostics of Black-hole Spin* (methods letter)
- *Schwarzschild Light Bending and X-ray Polarization for Cyg X-1*

Technical Skills

Python (NumPy, SciPy, Pandas, Matplotlib); Jupyter; L^AT_EX/Overleaf; Git; IDL; ADQL/Gaia; AstroImageJ/ASTAP, Fortran, and C++.

List of References

Prof. James M. Lattimer,

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Prof. Joshua Tan,

Department of Natural Sciences, LaGuardia Community College (CUNY);

Research Associate, American Museum of Natural History (AMNH)

New York, NY, USA

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